The Land Repair Helix

Site plan: Farm

Site name: _____ Date: _____ Prepared by: _____

Compacted, tired soil → Feed the soil life (Soil)

Compacted soil with little organic matter holds less water, drains badly, and needs more fertiliser to grow anything.

- ☐ £ Stop heavy machinery and livestock on wet ground
- ☐ £ Add compost, farmyard manure or digestate to build organic matter
- ☐ ££ Sow deep-rooting cover crops or herbal leys (chicory, plantain) to break up compaction
- ☐ ££ Cut down on ploughing; direct drilling where it suits the soil

NI: Soil sampling is often supported through DAERA schemes. Check what's open before paying for testing.

Heavy artificial fertiliser use → Let plants make the nitrogen (Soil)

Bagged nitrogen is costly and energy-intensive to make, gives off nitrous oxide, and runs off into water.

- ☐ £ Test soil first: correct the pH with lime before adding nutrients
- ☐ £ Plan nutrients from the analysis: put slurry and digestate where the crop needs them
- ☐ ££ Sow clover into grassland; legumes fix nitrogen from the air
- ☐ ££ Use low-emission slurry spreading to keep more nitrogen in the soil

NI: Under the Nutrients Action Programme, apply nutrients only where the crop needs them, and respect the closed periods and buffer distances from waterways.

Bare soil over winter → Keep it green (Soil)

Bare soil erodes in heavy rain and carries sediment and phosphorus into rivers and loughs.

- ☐ £ Leave stubbles, or sow a winter cover crop
- ☐ £ Keep grass margins at field edges and gateways
- ☐ ££ Plant hedges or grass strips across slopes to slow runoff

Flooding and fast runoff → Slow the flow (Water)

Hard surfaces and drained land send water downstream fast, causing floods and washing pollution off the land.

- ☐ £ Break up hard surfaces with gravel, grass-crete or permeable paving
- ☐ ££ Dig swales, rain gardens and small ponds to hold water back
- ☐ ££ Install leaky dams of logs or brash in ditches and small streams
- ☐ £££ Add sustainable drainage (SuDS) when sites are redeveloped

NI: Work in or near a watercourse may need consent from DfI Rivers. Check before building dams or culverts.

Dries out in summer → Hold water in (Water)

Dry soil cracks and stalls growth; plants and wildlife suffer in heatwaves.

- ☐ £ Build soil organic matter; healthier soil holds far more water
- ☐ £ Mulch beds and leave cut vegetation to protect the soil surface
- ☐ ££ Harvest rainwater from roofs into tanks for washing or irrigation
- ☐ ££ Restore ponds and wet hollows

NI: Taking more than a small amount of water from a river, lough or borehole may need an NIEA abstraction licence.

Blue-green algae (cyanobacteria) blooms → Starve the bloom of phosphorus (Water)

Too much phosphorus and nitrogen, plus warm still water, feeds toxic cyanobacteria. Lough Neagh is the clearest NI example.

- ☐ £ Test soil phosphorus and stop applying where the index is already high
- ☐ £ Fence livestock out of streams and provide drinking troughs
- ☐ £ Get septic tanks emptied and checked
- ☐ ££ Leave wide, ungrazed buffer strips of grass and trees along watercourses
- ☐ £££ Build constructed wetlands or reedbeds to catch nutrients from yards and drains

NI: Keep slurry and fertiliser off land next to waterways, and inside the buffer distances, especially in winter and before forecast rain.

Take care: Don't touch bloom scums, and keep dogs out of the water. Cyanobacteria can be toxic.

Diesel, oil or paraffin spilled on land → Contain, soak up, dispose properly (Spills)

A few litres of oil can pollute a large area of soil, a drain or a stream.

- ☐ £ Stop the leak and block nearby drains with covers or sandbags
- ☐ £ Soak it up with sawdust, wood shavings, straw, coir, cork granules or wool
- ☐ £ Clay granules (cat litter) work well on hard surfaces
- ☐ £ Keep a ready-made spill kit beside every fuel store and generator
- ☐ ££ Dig out badly soaked soil, then let microbes treat it in a biopile

NI: If oil reaches a drain, watercourse or groundwater, ring the NIEA Incident Hotline on 0800 80 70 60. Used sorbents are hazardous waste and must go via a registered carrier.
Take care: Don't use peat, which destroys bogs, or washing-up liquid and detergents, which spread the oil into the water.

Gappy, flailed hedgerows → Lay, gap up and let them grow (Rewilding)

Hedges cut hard every year flower little, give poor shelter and slowly die out at the base.

- ☐ £ Cut in rotation every 2 to 3 years, in winter, never during the closed period
- ☐ £ Let some hedgerow trees grow up
- ☐ ££ Plant gaps with native hawthorn, blackthorn, hazel and holly
- ☐ ££ Lay overgrown hedges to thicken them at the base

NI: In NI, hedge cutting is not allowed from 1 March to 31 August, except for clear road-safety reasons.

Few pollinators → Flowers from spring to autumn (Rewilding)

Bees and hoverflies need food from March to October, and nesting spots.

- ☐ £ Leave dandelions and clover in amenity grass
- ☐ £ Plant spring-flowering willow, hawthorn and blackthorn, and late-flowering ivy
- ☐ £ Leave bare south-facing banks for mining bees
- ☐ £ Cut pesticide use

NI: The All-Ireland Pollinator Plan has free guides for farms, councils, businesses and schools.

Ammonia from slurry and livestock → Cover, inject, and time it (Air and carbon)

Ammonia harms nearby woodlands, bogs and rivers, and forms fine particles that damage health. NI's ammonia emissions are mostly from agriculture.

- ☐ £ Spread in cool, damp, still weather, never in hot sun
- ☐ ££ Use low-emission spreading equipment such as trailing shoe or dribble bar
- ☐ ££ Cover slurry stores
- ☐ ££ Plant tree shelterbelts around sheds to catch ammonia

NI: DAERA's ammonia measures and the Nutrients Action Programme set rules on spreading equipment and timing.

Drained or damaged peat → Rewet and protect (Air and carbon)

Drained peat releases carbon dioxide year after year. Intact wet bog stores carbon and water.

- ☐ £ Never use peat-based compost or peat as a sorbent
- ☐ ££ Block drains on peat to raise the water table
- ☐ ££ Avoid planting trees on deep peat

NI: NI's peatland strategy aims to protect and restore bogs.

Trees in the wrong place → Right tree, right place (Air and carbon)

Trees planted on species-rich grassland, peat or nesting habitat can do more harm than good.

- ☐ £ Plant native broadleaves along watercourses for shade; cooler water helps reduce algal blooms
- ☐ £ Plant hedgerow trees and shelterbelts rather than blocks on good habitat
- ☐ ££ Let woodland regenerate naturally where there are seed sources

NI: Check for priority habitat and protected sites before planting.

Misconnected drains → Right pipe, right place (Water)

Washing machines, sinks or yard drains plumbed into surface water drains send dirty water straight to rivers.

- ☐ £ Check where every drain on your site goes: surface water or foul sewer
- ☐ £ Dye-test suspect connections
- ☐ ££ Reconnect misconnected appliances to the foul sewer
- ☐ ££ Fit oil interceptors on yard drains where vehicles park or refuel

NI: Discharging trade effluent to the foul sewer needs NI Water consent. Polluting a waterway is an offence.

Low soil carbon → Build carbon back in (Soil)

Soils low in carbon hold less water and fewer nutrients, and lose their structure.

- ☐ £ Keep the soil covered and growing for as much of the year as possible
- ☐ ££ Add compost and well-made digestate
- ☐ ££ Trial biochar from clean wood waste, mixed with compost
- ☐ ££ Turn some arable ground into grassland or herbal leys

NI: Spreading biochar on land may need a waste exemption or permit, depending on what it's made from.
Take care: Biochar quality varies. Buy certified material and don't expect results in one season.

Important. For contaminated land, forever chemicals (PFAS) or any spill reaching a drain, watercourse or groundwater, get specialist advice and involve the regulator. In NI, report pollution to the NIEA Incident Hotline on **0800 80 70 60** (24 hours). Contaminated-land remediation normally goes through planning, using the Land Contamination: Risk Management (LCRM) framework.
Sources. DAERA and CAFRE guidance on the Nutrients Action Programme and the hedge-cutting closed period (1 March to 31 August); NIEA incident reporting (nidirect); LCRM (Environment Agency, used by NIEA); UK and EU work on PFAS (Defra PFAS Plan 2026, ECHA); the All-Ireland Pollinator Plan; the "Don't Mow, Let It Grow" initiative run by NI councils and partners.
Position at 16 October 2026. For information, not legal or engineering advice. Robert J. Park · linkedin.com/in/robertparkuk · WhatsApp @robertjpark